

Pizza Sales Business Analysis Using SQL

Project Overview

Project Title

Pizza Sales Business Analysis Using SQL

Objective

The objective of this project is to analyze pizza sales data using SQL and answer real world business questions related to sales performance, customer ordering behavior, product performance, revenue generation, and sales trends. The project converts raw transactional data into meaningful business insights that support informed business decisions.

This project demonstrates practical SQL skills such as data retrieval, filtering, aggregation, joins, subqueries, Common Table Expressions (CTEs), and window functions. It also showcases the ability to solve business problems, calculate key performance indicators (KPIs), and present analytical findings in a structured and professional manner.

Dataset

The project uses four relational tables.

Table Name	Description
orders	Stores customer order details including order ID, order date, and order time.
order_details	Contains details of each pizza ordered, including pizza ID and quantity.
pizzas	Stores information about pizza size and price.
pizza_types	Contains pizza names, categories, and ingredients.

Tools Used

- MySQL Workbench
- SQL
- GitHub
- Microsoft Excel (for validating query results)

SQL Concepts Used

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- Aggregate Functions (COUNT, SUM, AVG, MAX)
- INNER JOIN

- CASE Statements
- Subqueries
- Common Table Expressions (CTEs)
- Window Functions
- Date and Time Functions
- Ranking Functions (RANK, DENSE_RANK)
- Percentage Calculations
- Cumulative Analysis

Project Scope

- Calculating the total number of customer orders.
- Calculating total revenue generated from pizza sales.
- Identifying the highest priced pizza.
- Finding the most commonly ordered pizza size.
- Identifying the top selling pizza types.
- Analyzing category wise sales performance.
- Identifying peak ordering hours.
- Calculating the average number of pizzas ordered per day.
- Measuring the revenue contribution of each pizza type.
- Analyzing cumulative revenue over time.
- Identifying the top performing pizzas within each category.

Expected Business Outcomes

The analysis performed in this project helps restaurant management to:

- Monitor overall business performance.
- Identify high performing pizzas and categories.
- Understand customer purchasing behavior.
- Improve inventory planning and stock management.
- Optimize staff scheduling during peak business hours.
- Support pricing and promotional strategies.
- Make informed business decisions based on data driven insights.